Software upgrade of broadband network gateways, Region East

Change window: 17–18 June 2027 · Risk: medium · Owner: IP Core team

1. Scope

This runbook describes the upgrade of the three broadband network gateways in Region East to software release 24.3.2. Each gateway is upgraded in its own one-hour window while its subscribers are carried by the redundant peer.

Site	Location	Active sessions	Window (UTC)
BNG-E01	Harlow exchange	18,420	00:30–01:30
BNG-E02	Chelmsford exchange	22,915	01:30–02:30
BNG-E03	Colchester exchange	15,077	02:30–03:30

2. Prerequisites

a) The change has been approved by the Change Advisory Board.
b) The new image has been copied to every gateway and its checksum verified.
c) The redundant peer of each gateway reports a healthy state.
d) The customer notice was published at least five days in advance.

3. Procedure

3.1 Pre-checks

Log in to the gateway from the operations jump host and record the current state before touching anything.

```
ssh noc-ops@bng-e03.mgmt.calderline.net
show version | include Software
show subscriber session summary
show redundancy group 1 status
copy running-config flash:/backup/pre-CHG-27-0613.cfg
```

Do not continue if the redundancy group reports anything other than a ready state.

3.2 Upgrade

The upgrade job reads its parameters from the following file, which is stored with the change record.

```
upgrade:
  target_release: "24.3.2"
  image: flash:/images/bng-24.3.2.bin
  checksum_sha256: 9f2c41e07ab35d8e
  drain:
    max_sessions_left: 50
    timeout_minutes: 20
  rollback_on_failure: true
```

1. Drain the gateway so that new sessions are steered to its peer.
2. Wait until fewer than 50 sessions remain, or 20 minutes have passed.
3. Install the new image and reload the gateway.
4. Confirm that the gateway rejoins the redundancy group.
5. Move sessions back by removing the drain.

3.3 Post-checks

Compare the session count with the value recorded in the pre-checks; a difference of more than five per cent must be investigated. Check that RADIUS accounting records are arriving for new sessions.

4. Rollback

If the gateway fails to rejoin its redundancy group within 15 minutes of the reload, boot it from the previous image. Rollback must be completed before the end of the window so that the next site can start on time.

5. Contacts

Role	Name	Phone
Change manager	Priya Nandakumar	+44 20 7946 0213
IP Core engineer on call	Tomasz Wielgosz	+44 20 7946 0388
Vendor support	Gateway TAC, contract 55-0917	+44 20 7946 0911

Escalate to the duty manager if neither engineer can be reached within ten minutes.